Heat Risk EU: Predicting Heat-Related Mortality and Mapping Urban Heat Vulnerability in Five European Countries

CAPSTONE PROJECT PROPOSAL

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1. Project Overview

Summer heat is the deadliest weather hazard in Europe, and summer 2026 made this very visible. According to provisional national estimates, Germany's Robert Koch Institute counted about 16,300 heat-related deaths by early September, Spain's MoMo surveillance system attributed 4,947 deaths to temperature since May, and Santé publique France recorded more than 7,000 excess deaths during the June and July heatwaves. These numbers come from different national methods, so they cannot be compared directly, and they say little about which neighbourhoods and which people carry most of the risk.

This project will build a data-driven heat risk tool for five countries along a climate gradient: Germany, France, Italy, Spain, and Greece. It will answer two practical questions. First, at what temperature does mortality start to rise in each region, and how does this threshold differ between a temperate city such as Berlin and a hot one such as Athens? Second, within a city, which neighbourhoods combine high surface temperature, little greenery, and many older residents, and therefore need cooling measures and heat warnings first?

The work happens at two levels. At the regional level, I will link Eurostat weekly death counts for every NUTS 3 region with population-weighted temperature from the Copernicus E-OBS dataset. A lagged quasi-Poisson regression, a simplified form of the distributed lag non-linear model widely used in heat epidemiology, will estimate each region's heat threshold and risk curve, and a Light GBM model will serve as a machine-learning comparison. Models will be trained on 2015–2025 and tested on the unseen summer of 2026. At the city level, I will build a Heat Vulnerability Index for Berlin, Paris, Rome, Madrid, and Athens on the 1 km European census grid, combining Landsat surface temperature, Sentinel-2 vegetation cover, Urban Atlas land use, and the share of residents aged 65 and over.

The project uses only open, official data. Pan-European datasets form the core, because they share one format across all five countries and make the comparison fair. National sources are used to validate the results: the model's 2026 heat-death estimates will be compared with the official figures from the Robert Koch Institute, Santé publique France, and Spain's MoMo system. I am not planning to collect new data or build a climate model; the work is to integrate existing sources reliably, model them carefully, and show the results in a way a non-specialist can use.

A Streamlit dashboard will let a user choose a country or city and view vulnerability maps, risk curves, and the factors driving risk. Engineering practices will make the work reproducible: Git version control, MLflow experiment tracking, pytest data checks, GitHub Actions, and a log of every data source version, since 2026 figures are still provisional. The goal is a validated, explainable prototype with clear evidence of how well it works, not only a heat map.

2. Proposed Solution and Methodology

2.1 Data sources and data preparation

The core pipeline relies on pan-European datasets with one consistent format. National datasets are used for validation and for checking local accuracy. All sources below were checked as publicly accessible in October 2026; direct links are listed in Appendix A.

Dataset	Provider	Use in this project
Deaths by week, sex and NUTS 3 region (with 20-year age groups)	Eurostat	Weekly mortality outcome for all five countries, 2015–2026
E-OBS daily gridded observations	Copernicus C3S / ECA&D	Daily mean and maximum temperature, 0.1° grid
ERA5-Land hourly reanalysis	Copernicus C3S	Humidity for apparent temperature (optional extension)
Census 2021 population grid (1 km)	Eurostat GISCO	Population weights and share aged 65 and over
Landsat 8/9 Collection 2 Level-2 Surface Temperature	USGS	Summer land surface temperature at 30 m for the five cities
Sentinel-2 Level-2A imagery	ESA / Copernicus Data Space Ecosystem	Vegetation index (NDVI) for the five cities
Urban Atlas 2021	Copernicus Land Monitoring Service	Land use, street tree layer, and building height
NUTS boundaries	Eurostat GISCO	Region shapes for joins and maps
Weekly heat mortality reports	Robert Koch Institute (DE)	Validation of German estimates
Heat and health surveillance data	Santé publique France (FR)	Validation of French estimates
MoMo daily mortality monitoring	Instituto de Salud Carlos III (ES)	Validation of Spanish estimates
Daily deaths by municipality	ISTAT (IT)	Finer-scale check for Italy (data to 30 June 2026)
National station observations	DWD, Météo-France, AEMET, NOA	Checking E-OBS temperature in the five cities

Before modelling, I will inspect each dataset for missing weeks, duplicated records, changed region codes, and unit differences (for example, Landsat surface temperature is delivered in Kelvin with a scale factor). All spatial data will be projected to one equal-area coordinate system (ETRS89-LAEA, EPSG:3035), all-time series aligned on ISO weeks, and all regions keyed to the NUTS version used by the Eurostat deaths table. A data log will record the source, version, and download date of every file, because the 2026 mortality figures are provisional and will be revised.

2.2 Heat exposure and vulnerability mapping

Regional heat exposure will be measured as population-weighted temperature: daily E-OBS values are averaged over each NUTS 3 region using the 1 km population grid as weights, so the measure reflects the heat where people live rather than over forests or farmland. Daily values will then be summarised into weekly mean and maximum temperature.

For the five cities, I will build a median summer surface temperature layer from cloud-free Landsat scenes taken in June to August 2022–2026, which reduces gaps from cloud cover and includes the 2026 heatwave. Vegetation (NDVI) from Sentinel-2, tree cover and building height from Urban Atlas, and the share of residents aged 65 and over will be aggregated to the same 1 km cells. The indicators will be standardised and combined into a Heat Vulnerability Index using principal component analysis to set the weights. Local Moran's I will then identify statistically significant clusters of high-vulnerability cells.

2.3 Temperature–mortality modelling

The baseline model is a quasi-Poisson regression of weekly deaths on temperature, with natural-spline terms for the current week and up to two previous weeks, plus seasonality, long-term trend, and region effects. This captures the non-linear, delayed effect of heat that the epidemiological literature describes. From the fitted curves I will derive, for each region, the minimum-mortality temperature, the threshold above which deaths rise, and the estimated number of heat-attributable deaths.

A LightGBM model with lagged temperature, week of year, region, and age-structure features will be trained as a comparison, to test whether a flexible machine-learning model predicts better than the established statistical approach. The COVID-19 years 2020–2022 distort mortality; I will handle them with indicator variables and check results in a sensitivity analysis that excludes those weeks. Training uses 2015–2025 and testing uses the summer 2026 weeks available at the time of download.

2.4 Evaluation and validation

Prediction quality will be measured on the held-out 2026 summer weeks with mean absolute error and root mean squared error, comparing the quasi-Poisson model, LightGBM, and a naive baseline (the average of the same week over the previous five years). Heat-attributable deaths will then be aggregated to federal state, department, or country level and compared with the official estimates from the Robert Koch Institute, Santé publique France, and MoMo. Because each agency uses its own method, this comparison focuses on whether the direction and magnitude agree, not on exact matches.

For the vulnerability index, I will check how stable the neighbourhood rankings are under different indicator weights, and compare the Berlin results with the independent climate analysis maps in the Berlin Environmental Atlas. SHAP values will explain which factors drive each prediction. I will establish baseline scores first and set numerical targets once those are known.

2.5 Engineering practices, testing, and deployment

Git and GitHub will hold all code and documentation. Pytest will cover data checks such as complete week sequences, valid region codes, and plausible temperature ranges, as well as key processing functions. GitHub Actions will run these tests automatically on each push. MLflow will record model settings and results, so the final configuration is chosen from evidence rather than guesswork.

The dashboard will read small, precomputed result tables rather than raw satellite data, which keeps it light enough to host. The first target is a reproducible local application; a public demo on Streamlit Community Cloud will be added if hosting limits allow. If it does not, the handover will include setup instructions and the commands needed to rebuild the results. Docker packaging is optional.

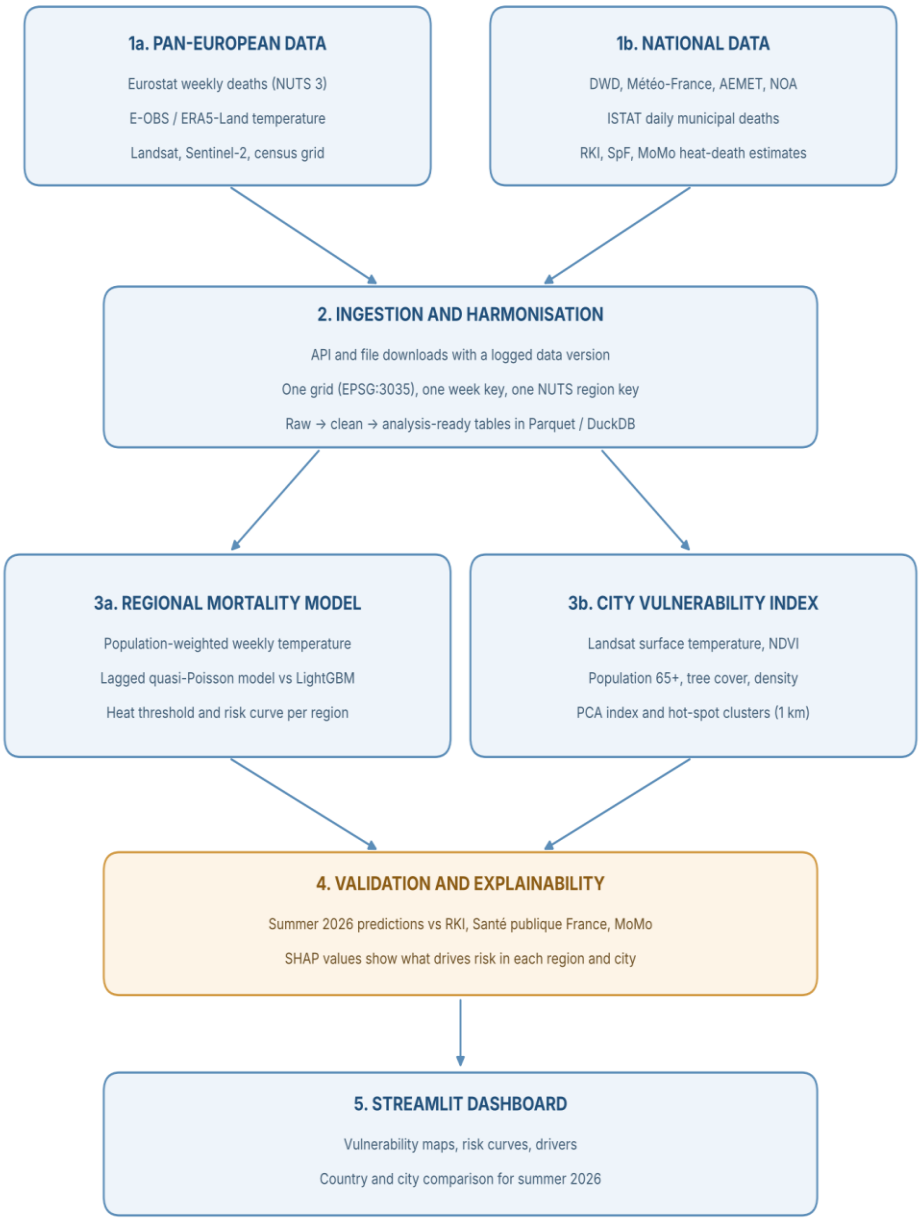
3. Technical Stack

The following tools are planned. All are free and open source; optional components will be used only if they fit the schedule.

- **Python:** Core language for the pipeline, modelling, and dashboard
- **pandas, DuckDB, Parquet:** Tabular processing and fast local storage
- **cdsapi, eurostat, pystac-client:** Programmatic download from Copernicus, Eurostat, and satellite catalogues
- **xarray, rioxarray, rasterio:** Handling gridded climate and satellite data
- **GeoPandas:** Spatial joins between grids, regions, and city boundaries
- **statsmodels:** Quasi-Poisson regression with lagged temperature splines
- **LightGBM, scikit-learn:** Machine-learning comparison model and evaluation metrics
- **SHAP:** Explaining model predictions
- **PySAL (esda):** Spatial hot-spot detection with Local Moran's I
- **MLflow:** Tracking experiment parameters and results
- **Streamlit, Plotly, Folium:** Interactive dashboard, charts, and maps
- **pytest, Git / GitHub, GitHub Actions:** Testing, version control, and continuous integration
- **Docker (if feasible):** Packaging for reproducible execution

4. System Diagram

The diagram shows the path from the data sources to the dashboard. The regional model and the city index are built in parallel from the same harmonised data, and engineering practices run across the whole workflow rather than as a final step.



5. Task Breakdown and Biweekly Sprint Plan

Each sprint ends with a progress checkpoint on the dates below. Work may overlap slightly between sprints as components are integrated.

Sprint / date	Focus	Planned tasks	Checkpoint deliverable
S1 10 November 2026	Data access and pipeline foundation	• Set up the repository, Python environment, and accounts (Copernicus Climate Data Store, USGS EarthExplorer, Copernicus Data Space). • Download Eurostat weekly deaths, NUTS boundaries, E-OBS temperature, and the 1 km census grid for the five countries. • Document schemas, missing weeks, and region-code changes; start the data version log. • Compute population-weighted weekly temperature for every NUTS 3 region. • Write the first pytest data checks.	Clean weekly deaths-and-temperature panel for all NUTS 3 regions, 2015–2026.
S2 24 November 2026	Regional mortality model	• Build the expected-deaths baseline with seasonality and trend. • Fit the lagged quasi-Poisson model; derive heat thresholds and risk curves per region. • Train the LightGBM comparison model and the naive baseline. • Handle the COVID-19 years and run the sensitivity check. • Record all runs and settings in MLflow.	Heat threshold and risk curve per region, with a first model comparison.
S3 8 December 2026	City vulnerability and validation	• Build median summer surface temperature layers from Landsat for the five cities. • Add Sentinel-2 vegetation, Urban Atlas tree cover and building height, and the share aged 65+. • Combine indicators into the Heat Vulnerability Index and detect hot-spot clusters. • Compare 2026 heat-death estimates with RKI, Santé publique France, and MoMo; add SHAP explanations. • Check the Berlin index against the Environmental Atlas climate maps.	Five city vulnerability maps and a validation report.

Sprint / date	Focus	Planned tasks	Checkpoint deliverable
S4 22 December 2026	Dashboard and final handover	• Build the Streamlit dashboard: maps, risk curves, drivers, and country comparison. • Configure GitHub Actions and complete test coverage. • Deploy to Streamlit Community Cloud if feasible; otherwise finalise local setup instructions. • Document limitations, data versions, and the final configuration. • Prepare the report, presentation, and demonstration.	Documented, presentation-ready dashboard and final report.

6. Gantt Chart

This high-level schedule shows the main work period for each activity. Sprints are checkpoints rather than isolated blocks, so testing and integration continue across sprint boundaries.

Activity	S1 10 Nov	S2 24 Nov	S3 8 Dec	S4 22 Dec
Setup and data access	●			
Data pipeline and harmonisation	●	○		
Regional mortality model		●	○	
MLflow experiment tracking		●	○	
Satellite processing		○	●	
Vulnerability index and hot spots			●	
Validation and SHAP explanations			●	○
Tests and GitHub Actions	○	○	○	●
Streamlit dashboard			○	●
Final report and presentation			○	●

7. Expected Outcomes and Risk Management

7.1 Expected outcomes

- A reproducible pipeline that rebuilds every dataset from the original public sources.
- A heat threshold, risk curve, and heat-attributable death estimate for every NUTS 3 region in the five countries and A comparison of the statistical model, LightGBM, and a naive baseline on unseen 2026 data.
- Heat Vulnerability Index maps for Berlin, Paris, Rome, Madrid, and Athens, with hot-spot clusters and A validation of 2026 estimates against official German, French, and Spanish figures.
- An interactive Streamlit dashboard, experiment records, automated tests, and continuous-integration checks.

7.2 How I will judge success

The project will be successful if the pipeline runs end to end from the public sources, the models produce reproducible results on the held-out 2026 data, and the dashboard shows the maps and risk curves clearly enough for a non-specialist to identify the most at-risk districts of a city. The model should at least match the naive baseline, the 2026 estimates should agree in direction and rough magnitude with the official figures, and I should be able to explain every modelling choice from the experiment records. Failure cases and limitations will be documented rather than hidden.

7.3 Risks and practical responses

Risk	How I will address it
Provisional or incomplete 2026 mortality data	Use the latest weeks available, log the data version, and rerun the validation once before submission. If a country's summer 2026 weeks are missing, test on summer 2025 instead and state this.
Different national methods for heat-death estimates	Compare countries only through this project's own harmonised model; use official figures for direction-and-magnitude checks within each country.
Cloud cover and gaps in satellite data	Build median summer composites from all cloud-free scenes in 2022–2026 rather than relying on a single date.
Processing limits on free cloud tiers	Limit satellite work to the five city areas; use USGS EarthExplorer downloads as a fallback to Copernicus Data Space.
COVID-19 distortion of 2020–2022 mortality	Add indicator variables and run a sensitivity analysis that excludes those weeks.
Area-level results misread as individual risk	State clearly that results describe places, not individuals, and avoid labelling specific neighbourhoods in a stigmatising way.
Limited time for an individual project	Prioritise the regional model, the five city maps, and the dashboard. Treat apparent temperature, national station checks, public hosting, and Docker as optional.

-Data Source Links

All links were checked as live in October 2026. Data access terms can change, so they will be re-checked at the start of Sprint 1.

Source	Link
Eurostat weekly deaths by NUTS 3 region	https://ec.europa.eu/eurostat/databrowser/view/demo_r_mwk3_t/default/table?lang=en
E-OBS gridded observations (Copernicus CDS)	https://cds.climate.copernicus.eu/datasets/insitu-gridded-observations-europe
ERA5-Land hourly reanalysis	https://doi.org/10.24381/cds.e2161bac
Eurostat Census 2021 population grid	https://ec.europa.eu/eurostat/en/web/gisco/geodata/population-distribution/population-grids
Landsat Collection 2 Level-2 products (USGS)	https://www.usgs.gov/land-resources/nli/landsat/landsat-collection-2-level-2-science-products
Copernicus Data Space Ecosystem (Sentinel-2)	https://dataspace.copernicus.eu
Copernicus Urban Atlas 2021	https://land.copernicus.eu/en/news/urban-atlas-2021-product-update-is-now-available
Robert Koch Institute (heat mortality reports)	https://www.rki.de
Santé publique France heat and health data	https://www.santepubliquefrance.fr/fortes-chaleurs-canicule/donnees
ISCIII MoMo mortality monitoring	https://www.sanidad.gob.es/va/areas/sanidadAmbiental/riesgosAmbientales/frioExtremo/momoBajasTemperaturas.htm
ISTAT daily deaths by municipality	https://www.istat.it/it/archivio/240401
DWD Climate Data Center	https://opendata.dwd.de/climate_environment/CDC/
Météo-France open data	https://meteo.data.gouv.fr
AEMET OpenData	https://opendata.aemet.es/centrodedescargas/productosAEMET
NOA meteo.gr station data (CLIMPACT)	https://data.climpact.gr/en/dataset/497dc26d-45e0-4ad5-b8f3-5f8890f65129
Berlin Environmental Atlas, Climate Analysis 2022	https://www.berlin.de/umweltatlas/klima/klimaanalyse/2022/download/